

Marketing Expenditure & Sales Analysis

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1 Introduction

This document demonstrates how to import data from a `.csv` file, perform a statistical analysis, and present findings using **RMarkdown**.

2 Data Import & Preparation

First, import sales data from the file named `marketing_data.csv`, located in the `data` subdirectory, using `read_csv()`. Then, preview top records to be sure they look as you expect:

```
## # A tibble: 6 x 4
##   Store_ID Ad_Budget_1k Sales_1k Region
##     <dbl>      <dbl>    <dbl> <chr>
## 1      101         2.5     15.2 North
## 2      102         3.1     18.5 North
## 3      103         1.8      11  South
## 4      104         4.5     25.4 East
## 5      105         5      27.8 West
## 6      106         2.1     13.4 South
```

3 Statistical Analysis

3.1 Summary Statistics

Next, compute basic descriptive statistics for ad expenditure and sales, across regions.

Table 1: Summary Statistics of Marketing Data (\$1,000s)

Avg_Ad_Budget	SD_Ad_Budget	Avg_Sales	SD_Sales
3.375	1.178073	19.475	5.986831

3.2 Regression Modeling

Fit a simple linear regression model to quantify how marketing expenditure impacts sales:

$$\text{Sales} = \beta_0 + \beta_1(\text{Ad_Budget}) + \epsilon$$

```
##
## Call:
## lm(formula = Sales_1k ~ Ad_Budget_1k, data = marketing_data)
##
```

```
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.6302 -0.2410  0.1234  0.2627  0.4195
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    2.3603     0.4812   4.905  0.0027 **
## Ad_Budget_1k    5.0710     0.1355  37.411 2.43e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4225 on 6 degrees of freedom
## Multiple R-squared:  0.9957, Adjusted R-squared:  0.995
## F-statistic: 1400 on 1 and 6 DF, p-value: 2.435e-08
```

3.2.1 A quest to improve the way the output appears

You might notice that the output is not very neat. This is the ‘standard’ output given from the `lm()` function. Newer packages can extract the information from any `lm` function output - here the `broom` package functions `tidy` and `glance` are used, together with the `kable` function, to obtain improved formatting of the same output.

Table 2: Estimated coefficients from linear model fit to marketing data.

term	estimate	std.error	statistic	p.value
(Intercept)	2.3603	0.4812	4.9045	0.0027
Ad_Budget_1k	5.0710	0.1355	37.4110	0.0000

Table 3: Model fit statistics, from estimated linear model.

r.squared	adj.r.squared	sigma	statistic	p.value	df	df.residual
0.9957	0.995	0.4225	1399.583	0	1	6

With that done, let’s get back to understanding the fitted model!

4 Visualization

Next, plot the data together with the estimated linear model. It uses the `ggplot2` package, which is included with the `tidyverse` package.

5 Discussion & Conclusions

- **Relationship Strength:** The simple linear regression indicates a strong positive association between advertising spend and store revenue.
- **Effect Size:** The slope estimate ($\beta_1 \approx 5.2$) suggests that each additional \$1,000 spend in advertising generates approximately (‘on-average’) \$5,200 in revenue.
- **Next Steps:** Expand dataset across additional quarters and run a multiple regression including regional fixed effects.

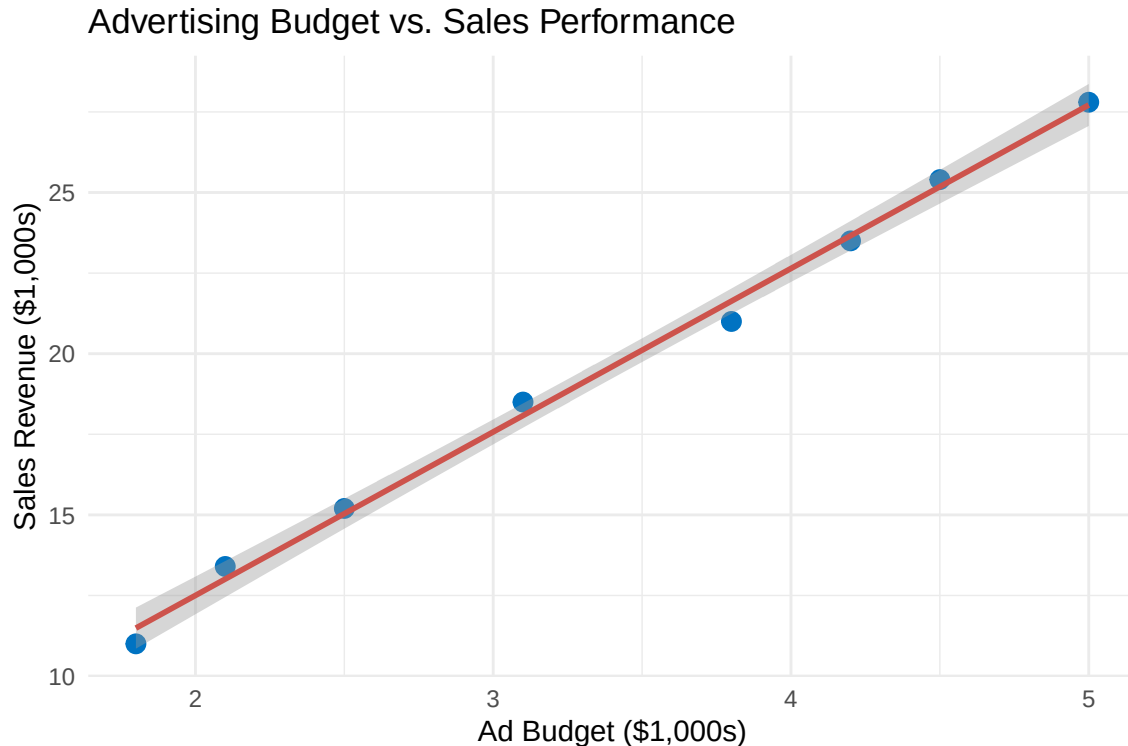


Figure 1: Scatter plot of Ad Budget vs. Sales Revenue

6 Additional Notes

1. You will need to install the `knitr`, `tidyverse` and `broom` packages (if not already installed) before running this code. You can find all of these packages on CRAN, and install them via the **Packages** tab.
2. The section title (“1 Introduction”) appears in larger font and in bold. This is the default setting, and is achieved by typing `#` before the word “Introduction” in the .Rmd file.
3. The subsection title (“3.1 Summary Statistics”) appears in large, bold font, though it is not as large as a section title. This is achieved by typing `##` before the words “Summary Statistics” in the .Rmd file.
4. The word “RMarkdown” above is shown in bold font. This was achieved by inserting two asterisks (`**`) before and after the word “RMarkdown”.
5. The table printed under 2. Data Import & Preparation was produced by simply asking R to print the relevant data frame. But this is not very ‘pretty’. A better option is to use the `kable` function, which was used to plot Table 1 in Section 3.1, for example.
6. To compile this .Rmd file, click on the Knit icon, and select **Knit to pdf**. You can also try the other output files, i.e. **Knit to HTML** (can be viewed in, e.g., a Google Chrome browser), or **Knit to Word**. (It may be best to avoid **Knit to ioslides_presentation**, as we haven’t added any formatting for this type of output.)